Biol. Cybern. 41, 19–32 (1981)

Biological Cybernetics

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Simulation of Head Movement Trajectories: Model and Fit to Main Sequence

W. H. Zangemeister[1], S. Lehman, and L. Stark

Departments of Engineering Science, Physiological Optics and Neurology, University of California, Berkeley, CA 94720, USA

Abstract. A sixth order nonlinear model for horizontal head rotations in humans is presented and investigated using experimental results on head movement trajectories and neck muscle EMG. The controller signals, structured in accordance with time optimal control theory, are parameterized, and controller signal parameter variations show a dominating influence on different aspects of the head movement trajectory. The model fits the common head acceleration types over a wide range of amplitudes, and also less common (dynamic overshoot) trajectories.

Introduction

Head movements are important in gaze being synkinetic with eye movements although generally the head moves later than the eye. There is also the important compensatory eye movement, driven by the vestibulo-ocular reflex in the direction opposite to head movement. Some properties of head movements are similar to eye movements. However, we would expect some differences both because of the differing mechanical loads, and also because of the relative prominence of the stretch reflexes in head movements and their functional absence in eye movements (Abrahams, 1977; Dichgans et al., 1974; Cooper et al., 1949; Peterson, 1979).

In addition to carefully observing and quantifying dynamic trajectories and the electromyographic controller signals, which drive these trajectories, it is most instructive to compare and contrast the experimental data with a mathematical model.

Many physiologists have studied eye muscles, and conceptual modelling of the eyeball and its muscles is by no means new. Descartes used two opposing eye muscles to illustrate his theory of reciprocal innervation as early as 1626. Mathematical modelling of the eye and its muscles *is* relatively new. Westheimer (1954) developed a second order linear model and Robinson (1964) a fifth order linear model: both models reproduced experimental displacements but not saccadic velocities or accelerations; they neither reflected eye nor eye muscle physiology explicitly, that is in an homeomorphic fashion in the sense of Bellman (1957). Cook and Stark (1968), Clark and Stark (1975), Hsu et al. (1976), and Lehman and Stark (1980) were able to subsequently explore and improve a sixth order nonlinear model of the eye plant, using reciprocally innervated pulse step controller signals to fit velocities and accelerations of experimental saccades and further showed that this model could reproduce smooth pursuit and vergence eye movements. To aid the exploration and further development of the model, Hsu et al. (1976), and lately Lehman and Stark (1980) used analyses of sensitivities of model parameters. Since models have now become more advanced and realistic they have been applied to questions of physiology (Bahill and Stark, 1975; Bahill and Stark, 1975; Young and Stark, 1963; Raphan and Cohen, 1977; Bock and Zangemeister, 1979) and clinical syndromes (Selhorst et al., 1977; Zee et al., 1976; Zangemeister and Bock, 1979).

Recently, some effort has also been undertaken to simulate various head movement trajectories. Besides vertical and tilt movements (Viviani et al., 1975; Walker et al., 1973; Reber et al., 1979), horizontal head rotations and their mechanics have been described by second order linear models in monkey (Bizzi et al., 1976) and humans (Morasso et al., 1977; Shirachi et al., 1978; Sugie et al., 1970; Meiry, 1971). A nonlinear

[1] On leave from Department of Neurology, University of Hamburg, F.R.G.; supported by Deutsche Forschungsgemeinschaft Bonn, FRG

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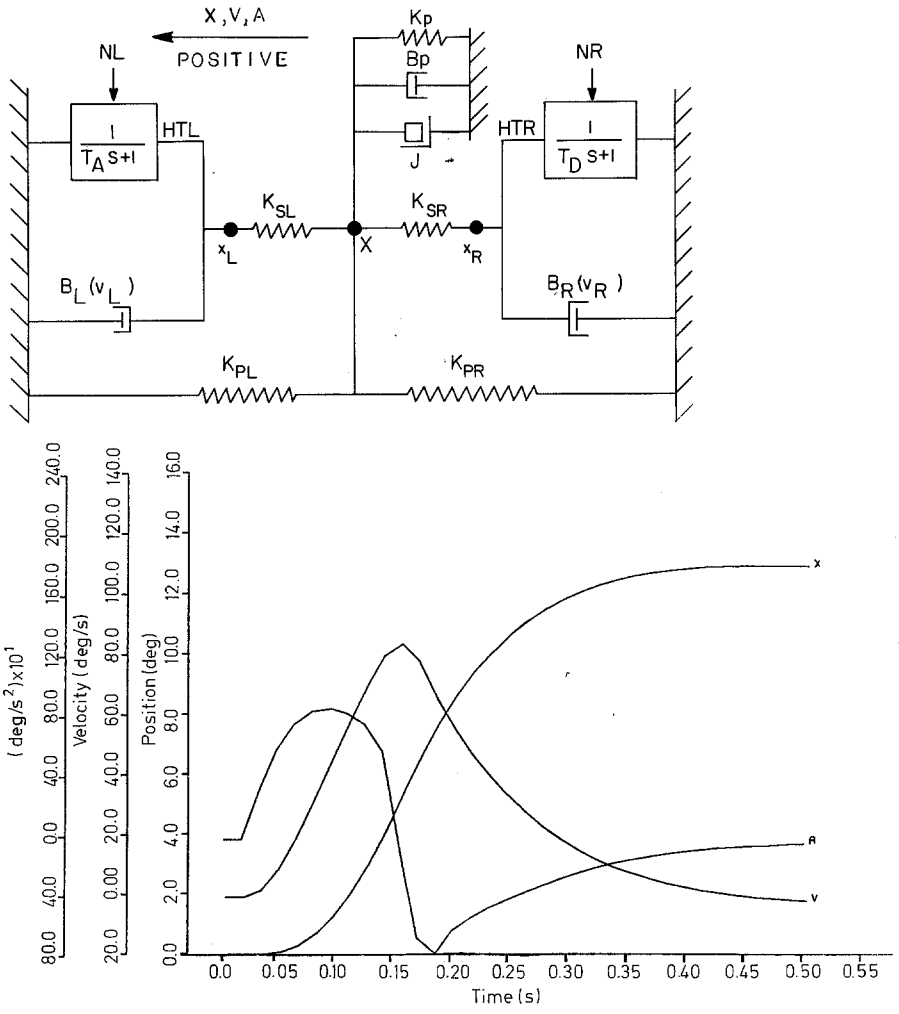


Fig. 1a. Structure of head movement model. K_p, B_p, J: parallel elasticity, viscosity and inertia of the head; K_{SL}, K_{SR}, B_L, B_R: left and right series elasticities and viscosities respectively; T_A, T_D: time constants of activation and deactivation. HTL, HTR: hypothetical tensions, left and right. NL, NR: neural control signal left and right; X, V, A: position, velocity, acceleration. **b** Model output for a 13 degree movement, axes as indicated, X, V, A: position, velocity, acceleration

homeomorphic model for horizontal head rotations has not been put forward before the present paper.

The theory of optimality has tempted many theorists with biological interests. A notable overall view is that of Rosen (1967). The closest study to our interests was that of muscle by Fitzlugh (1977), who applied Pontryagin's Maximum Principle to a smaller system. Also of interest are the papers by Bogumil (1980), Smith (1962), and Cook et al. (1968, 1975, 1980).

Although many types of head movements are possible given the wide variety of controller signals and shapes that the brain can use to generate these movements, it is possible by careful instructions to cooperating subjects to obtain as fast as possible, or time optimal, head movements. In this special case optimally theory (Feldbaum, 1965; Intrilligator, 1971; Bellman, 1957; Pontryagin, 1962; Rosen, 1967) can be used to both predict the type of controller signal which would produce an optimal trajectory and also deduce characteristics of this optimal trajectory (Lehman and Stark, 1980; Zangemeister et al., 1980). Given these idealized signals and trajectories, one can then study non-time optimal head movements as divergences from the ideal.

The present study defines a sixth order nonlinear model for horizontal head rotations to aid in the heuristic process of both direct and indirect modelling. Patterns of parameters of this model are explored, supported by such important tools as sensitivity analysis (Tomovic and Vucobratovic, 1972) and the Bremermann Optimizer (Bremermann, 1970). Various control signal patterns and their influence on the model's dynamical behavior are tested, and are then compared and contrasted with experimental results on electromyographic evidence for neurological control of head movements [Zangemeister et al. (1980) and the dependent dynamical features of head rotation (Zangemeister et al., 1981)]. In particular, the model closely fits both common and special acceleration trajectory types – as related to the underlying control signal – and also fits experimental head movement main sequence data. These findings are then compared with earlier simulations of eye (Clark et al., 1975; Lehman and Stark, 1980) and arm (Stark, 1968)

movements and their underlying control signals. Our model, besides giving another example of the utility of homeomorphic models in biomedical research, also provides additional background for understanding normal and clinically abnormal head and eye movements and their coordination.

Methods

1. Model Structure

The structure of our model is represented in Fig. 1a. It consists of two symmetrical muscles – representing the horizontal head rotating muscle complex in man (i.e. right and left mm. splenius and sternocleidomastoideus, plus others like semispinalis capitis and also smaller ones) acting synergistically for horizontal head movements – including contractile components, apparent viscosities, and elasticities, connected to a node representing the head, which has its own inertia, viscosity and elasticity.

Input to the model are innervations NL and NR in gram force equivalents. Although of course the inputs to neck muscles are actually the firing frequency of motor neurons, these have been converted into gram force, with no limiting dynamics from the synaptic junction at the motor endplate.

NL and NR pass through first order processes with time constants T_A and T_D, representing calcium activation of each muscle and possibly other biophysical processes. The resulting variables are hypothetical tensions HTL and HTR, which are then combined with the appropriate velocities in Hill's hyperbolic force-velocity relationship, to give equivalent dissipative nonlinear viscosities B_L and B_R, acting in parallel with the tension generators. Concavity of the hyperbola is determined by two constants: Hill's a and b; in our model, $b = Bh = 0.25$ times v_{max}, the maximum velocity of an unloaded muscle, and $a = \text{afact} = 0.25$ times P_0, the peak attainable force. Note the symmetry. In the diagram (Fig. 1a) x_L and x_R indicate the hypothetical positions of the nodes at which v_L and v_R, velocities of the force-velocity relationship are calculated. The parallel elasticities of both muscles can be lumped with the elasticity of the passive structures – that is everything but activated muscles – to yield a total parallel elasticity K_p. This lumping is justified since the elasticities act in parallel at the same node, whose position x, velocity v, and acceleration a define the external dynamics of head trajectories. Series elasticities of left and right muscles are denoted by K_{SL} and K_{SR}, respectively. The skull itself is modelled as a rigid sphere of radius 8.0 cm, with about the density of water, giving the inertia, labelled J. We were able to establish many of the model parameters by comparision with and scaling from the eye model (Clark, 1975; Lehman and Stark, 1980) and by using available

Table 1. Example of a numerical model output; parameters as indicated in Fig. 1a

```
*D EDTS
*:MODEL.LP
*:MMOSS.DA</7S
 PARAMETERS
J =
.18
BP =
2.
KP =
2.
KSL =
43.
NLMAX =
600.
TIME CONSTANT =
.05
TNL =
40.
TNR =
30.
TS,TS2,TS3 = (SECONDS)
.117,.147,.147
 TIMESP =
.350
BH =
150.
PLOT OUTPUT? 1 FOR NUMBERS,2 FOR PLOT
1
MASK OUTPUT TO -- SECONDS
-0.040
0.015      0.45779161E-02      0.29396885E+00      -0.35543965E+01
0.030      0.87117397E-02      0.24469327E+00      -0.33246531E+01
0.015      0.22523456E-01      0.25261121E+01       0.32744216E+03
0.030      0.10802292E+00      0.96456379E+01       0.66033559E+03
0.045      0.32657469E+00      0.20668281E+02       0.77295792E+03
0.060      0.71717953E+00      0.32373386E+02       0.85574139E+03
0.075      0.12959324E+01      0.45407134E+02       0.87190219E+03
0.090      0.20758561E+01      0.58265545E+02       0.83621188E+03
0.105      0.30432976E+01      0.73381020E+02       0.76427471E+03
0.120      0.41761577E+01      0.88955669E+02       0.65492709E+03
0.135      0.54336197E+01      0.84766747E+02      -0.74360571E+02
0.150      0.66695328E+01      0.78914391E+02      -0.66923819E+03
0.165      0.77657830E+01      0.66876542E+02      -0.76951999E+03
0.180      0.86906375E+01      0.56713086E+02      -0.68525584E+03
0.195      0.94772100E+01      0.48267352E+02      -0.52378524E+03
0.210      0.10145741E+02      0.40946577E+02      -0.45581691E+03
0.225      0.10711815E+02      0.34499031E+02      -0.40479981E+03
0.240      0.11154497E+02      0.28796739E+02      -0.35625362E+03
0.255      0.11579532E+02      0.23817993E+02      -0.33841193E+03
0.270      0.11904329E+02      0.19540685E+02      -0.25311063E+03
0.285      0.12159730E+02      0.15939564E+02      -0.22244232E+03
0.300      0.12384986E+02      0.12847471E+02      -0.19726502E+03
0.315      0.12558319E+02      0.10273486E+02      -0.15723855E+03
0.330      0.12695592E+02      0.81134271E+01      -0.13194291E+03
0.345      0.12803417E+02      0.63029733E+01      -0.11743966E+03
0.360      0.12896327E+02      0.47973657E+01      -0.92389595E+02
0.375      0.12943411E+02      0.35203742E+01      -0.77174770E+02
0.390      0.12993879E+02      0.24629461E+01      -0.64359432E+02
0.405      0.13023233E+02      0.15928672E+01      -0.53548145E+02
0.420      0.13041317E+02      0.84994953E+00      -0.44453927E+02
0.435      0.13049364E+02      0.24239355E+00      -0.36305673E+02
0.450      0.13049395E+02     -0.25219351E+00      -0.29351295E+02
0.465      0.13041847E+02     -0.69575225E+00      -0.25995163E+02
0.480      0.13029793E+02     -0.10391555E+01      -0.21335973E+02
   T             P                   V                   A
```

head dynamical data for humans (Reber et al., 1979; Walker et al., 1973; Viviani et al., 1975) and monkeys (Bizzi et al., 1976). The remaining parameters, particularly time constants and B_p, the viscosity, were estimated on physiological grounds, then tuned to fit model output to experimentally recorded trajectories. The viscosity, B_p, includes nonactive surrounding muscles, connective tissue, ligaments, and the atlanto-occipital joint. A set of six coupled ordinary differential equations, the state equations, describe the complete model

$$\dot{x} = v$$

$$\dot{x}_L = (\text{HTL} - K_{SL}(x_L - x))/B_L$$

$$\dot{x}_R = (-\text{HTR} - K_{SR}(x_R - x))/B_R$$

$$\dot{v} = (-K_p x - B_p v + K_{SL}(x_L - x) + K_{SR}(x_R - x))/J$$

$$\dot{\text{HTL}} = (1/T_A)(NL - \text{HTL})$$

$$\dot{\text{HTR}} = (1/T_D)(NR - \text{HTR}),$$

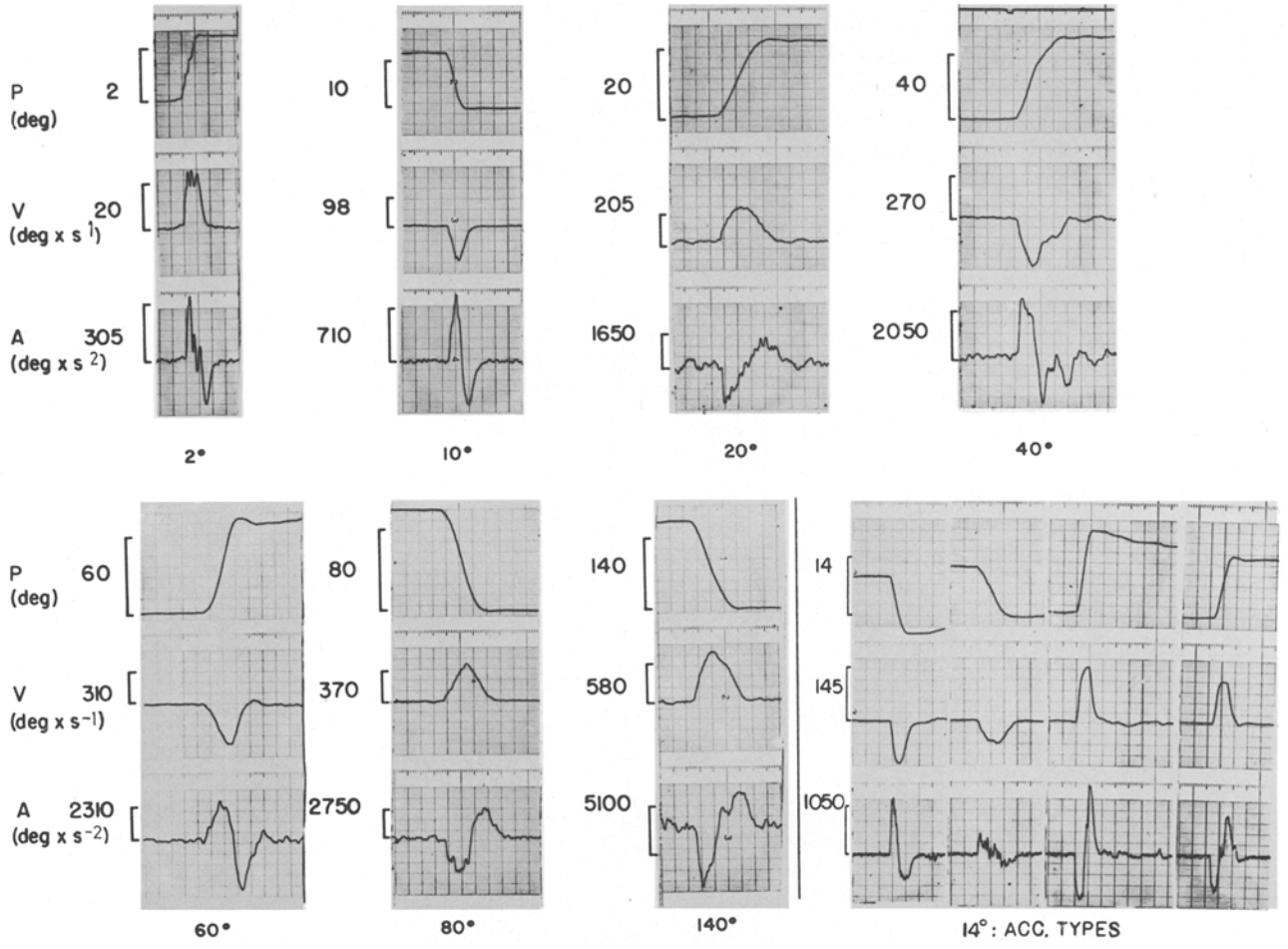


Fig. 2. Experimental head trajectories. *P*, *V*, *A*: position, velocity, acceleration for different amplitudes (2–140 degrees) and types (lower right hand part)

where B_L and B_R are given by the ancillary equations:

$$B_L = \begin{cases} (\mathrm{HTL} + \mathrm{afact}^{*}\mathrm{HTL})/(Bh + v_L) & v_L \geqq 0 \\ 3\mathrm{HTL}/Bh & v_L < 0 \end{cases}$$

$$B_R = \begin{cases} 3\mathrm{HTR}/Bh & v_R \geqq 0 \\ (\mathrm{HTR} + \mathrm{afact}^{*}\mathrm{HTR})/(Bh + v_R & v_R < 0. \end{cases}$$

Note that the nonlinearities of the model occur due to the presence of B_L and B_R in the denominators of the second and third state equations; B_L and B_R also contain the state variables HTL and HTR. A typical numerical output of a model simulation on a minicomputer (Fig. 1b) also lists the parameter values (Table 1). TNL and TNR stand for left and right tonic forces. $\mathrm{NL}_{\max}$ refers to the maxium "pulse height" value attainable by NL and NR, the innervations of the two muscle pairs. Since the input to the muscles comes from many individual neurons connecting to many muscle fibers, pulse height may be thought of as a measure of the number of these firing single neurons times the firing rate. As mentioned above this total innervation figure is converted into muscle force.

To give an example (Fig. 1b), we drove the model with 700 gram-force double pulse-step [agonist pulse width (sw1)] 0.0110 s, antagonist pulse width (sw2) 0.037 s (0.147–0.110 ms), tonic steps 40 (TNL) and 20 (TNR) gram-force (sw 3 is optionally used only for third order control), to find the relative timing of changes in acceleration ($ta_{\max}$ and $ta_{\max}$), velocity ($tv_{\max}$), and position (duration) due to a change in NL. Obviously, peak acceleration followed NL and NR rapidly, whereas peak velocity is later at 110 ms, and final head position is reached only after a relatively long time, 280 ms (compare with numerical output Table 1). Besides the ramplike position trace, other basic trajectory patterns can be seen: the high peaked monophasic velocity curve, and the biphasic acceleration trace, with maximum positive acceleration being greater than maximum negative acceleration. As an example of possible slight irregularities the deceleration part shows a second maximum.

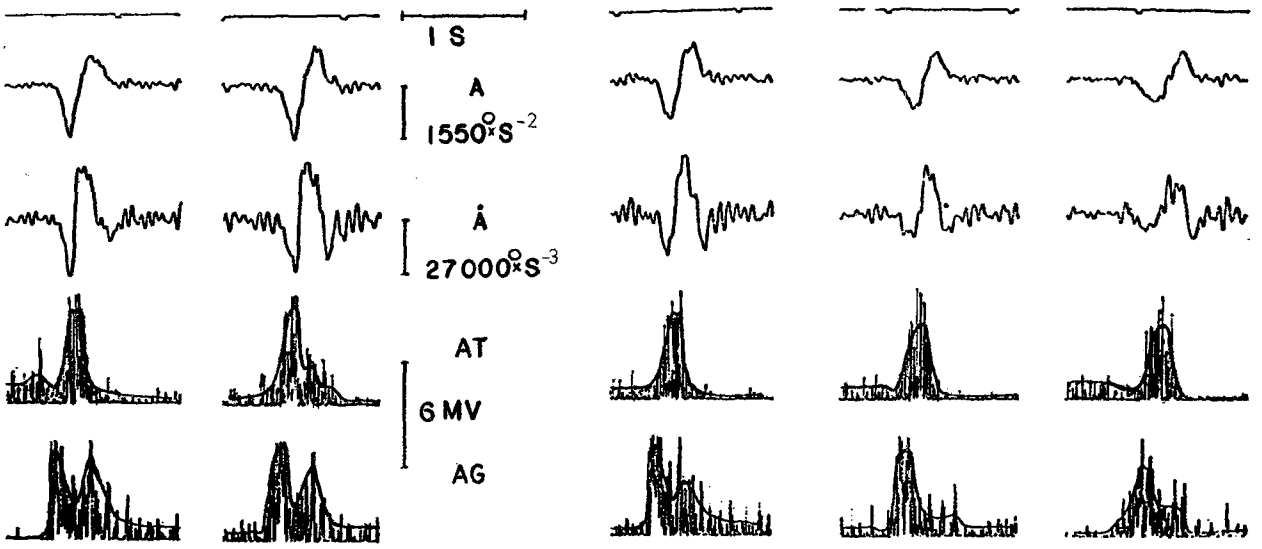


Fig. 3. Experimental head trajectories and rectified and integrated EMG of the splenius muscles. *A*, *A* dot: acceleration, first derivative of acceleration; *AT*, *AG*: agonist and antagonist EMG

2. Experimental Background

2.1. Main Sequence of Horizontal Head Rotation Trajectories. A set of experimental trajectories (Fig. 2) for a wide range of amplitudes on a normalized scale shows the significant characteristics of the dynamics of horizontal head rotations: Rapid ramplike changes of position, monophasic peaked velocities and biphasic acceleration curves – the latter showing usually higher peaks for positive accelerations. The acceleration and velocity curves as critical features of horizontal head rotations can be related to "bang-bang" double pulse control signals for these most rapid head rotations. Some special features can be noted, particularly in acceleration traces: sharper peaks in small (2 degree) movements compared to rounded maxima for large ones (80 degree); also, "fragmentation" of the curves occurs for both small (2 degree) and large (40 and 80 degrees movements – possibly related to irregular braking by antagonist, stretch reflex or fatique, shown in the movement's decomposition into perhaps dual movements (40 degree).

This leads to a second important qualitative observation on variability of trajectory dynamics, categorized by us previously into five typical acceleration patterns. One type shows sharply peaked acceleration traces (10 degree, Fig. 2) with only little difference between positive and negative accelerations, representing fairly symmetrical bang-bang control. A second type has much smaller maximum negative accelerations, with varying concave or convex shapes: these trajectories, the most common type, represent dynamical results of asymmetrical bang-bang controller signals. As expected the decelerating controller signals have a variety of shapes (Fig. 3, 14 degree left, 140 degree). A third type, more often seen in larger movements (60, 80 degree; 14 degree on very right), shows rounder acceleration curves, perhaps because of saturation phenomena. A fourth type has a greater and more peaked negative acceleration as compared to smaller and rounder positive acceleration (60 degree, 14 degree third from left) – thus showing inverse symmetry to the most common type. A fifth type has a typical "fragmented" acceleration trace (2, 20, 80, 140 degree and 14 degree second from left). This rapid oscillation at midpoint of trajectory apparently represents multiple switchings at mid-trajectory of agonist and antagonist muscle, or for just the agonist muscle alone. These irregularly fragmented trajectories are reminiscent of double saccades seen in fatigued humans (Bahill et al., 1975). Similarly, occasionally dynamic (60 degree) and glissadic (14 degree, third from left) overshoots are seen in head rotations, and are reminiscent of eye movements. Static overshoots are, of course, most common but since they are the result of a second discontinous neurological command signal, they are not variants of trajectory dynamics.

2.2. Correlation between Neck Muscle EMG *as a Sample of the Underlying Controller Signal and Head Movement Trajectories.* A significant aspect of the close, and causal, relationship between acceleration of horizontal head rotation and the underlying control signal, of wich the neck muscle electromyogram is a sample, is the steepness of initial acceleration part, when compared with the rapidity of increase of en-

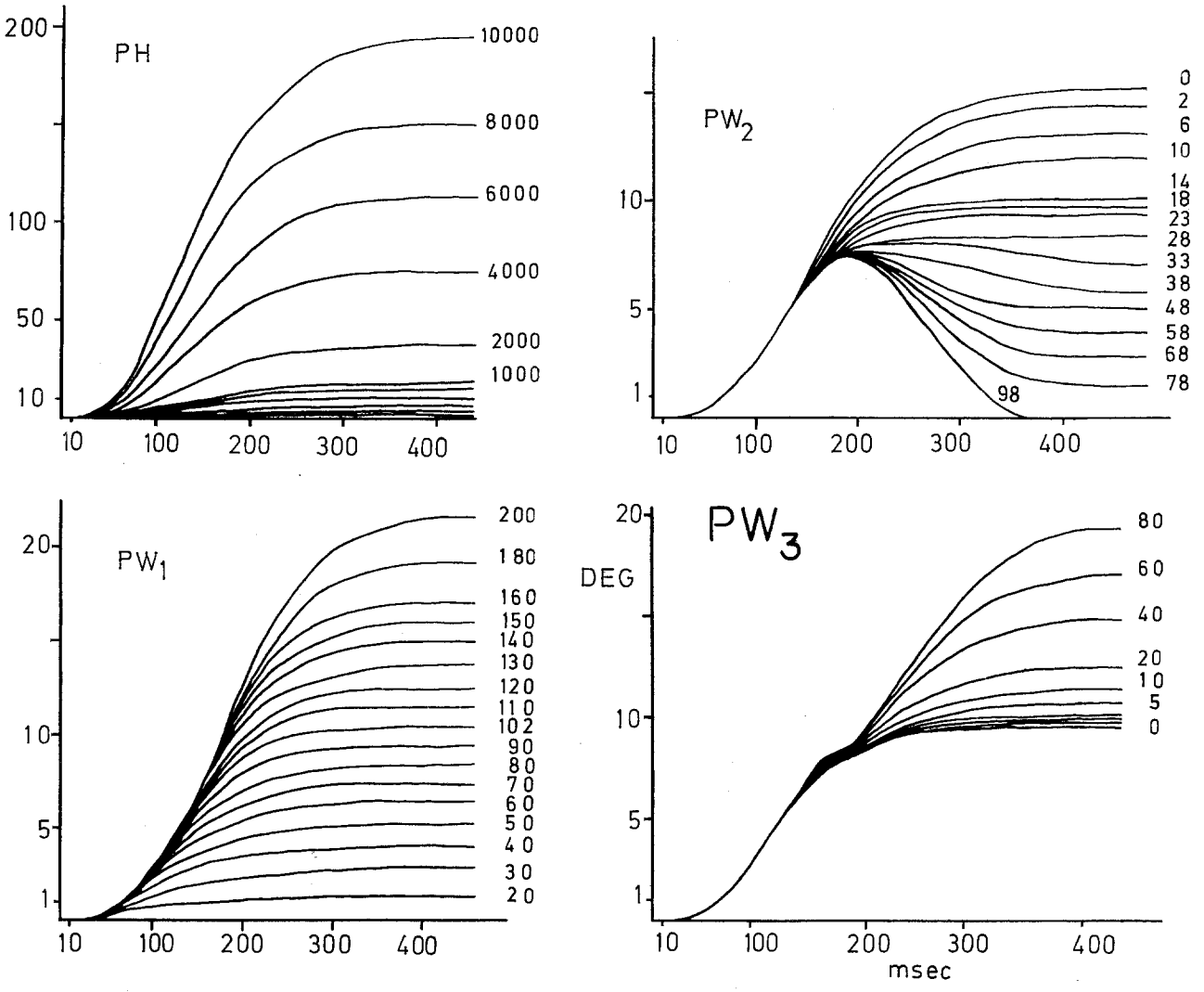


Fig. 4. Lawful variations of control parameters. PH, PW_1, PW_2, PW_3: pulse height, agonist pulse width 1, antagonist pulse width 2, second agonist pulse width 3; abscissa and time in ms; ordinate: amplitude in degree

velopes of agonistic and antagonistic EMG (Fig. 3, compare very left with very right movement of the same amplitude). This intimate correlation between EMG and the variable behavioral dynamics of head rotation is further reflected in amplitude of the first derivative of head acceleration. The bi- and tri-phasic nature of acceleration curves and their derivatives appear to match reciprocal innervation of the splenius muscles, shown in the rectified and smoothed EMGs: When the acceleration derivative's traces are multiphasic the EMG appears also to be multiphasic, particularly for fast, almost time optimal, movements (Fig. 3, three left movements); slow movements also show this tight correlation in spite of their different dynamics, (more biphasic, two right movements). This is a further demonstration of causal sequence from EMG to higher derivatives of head rotation, even in phases of control where subjects by varying intent choose from a wide selection of head movement patterns. This optional variety of head movements is in contrast to the stereotyped movements of the eye rotations as will be discussed below.

Results

1. Families of Controller Signal Variations

1.1. Pulse Height. As we learned from the sensitivity analysis the control parameters are the most influential ones in the model. We now give a more explicit idea of what that means. Pulse height strongly influences the amplitude of a particular head movement, given a constant agonistic and antagonistic pulse width (Fig. 4, upper left). Because of the sixth order nonlinear model the curve is more smoothly curved in its initial rising part; also this occurs with respect to the agonistic

pulse width, showing the dynamic delays of this higher order model. Although this cannot be generalized, this is true for our specific models. It shows how pulse height has a quasi linear influence, and how this controller signal parameter does not produce a time optimal response. By this we mean, that for a given amplitude the model response generated by varying pulse height is non time optimal when compared to a variation of agonistic pulse width producing the same amplitude (Fig. 4, left part).

1.2. Agonistic Pulse width (PW1). When pulse height is kept constant and the agonistic pulse width 1 is changed, this variation drives the particular head movement model responses with the time optimal type of control using the maximal available force. As agonistic pulse width becomes longer, the amplitude of the movement increases quasi linearly but by a smaller amount than resulted from pulse height influence. Also one can see how trajectory duration increases with this longer agonistic pulse duration. Note that the changes were between double and 1/10 of the initial starting values. Especially interesting is that for smaller head movements the final amplitude is reached much earlier than in head movements driven by different pulse heights. This of course is a rough version of the time optimality principle.

1.3. Antagonistic Pulse width (PW2). Antagonistic pulse width, which is the duration of the neurological reversal pulse, helps to terminate the early dynamical phase of the movement, and it dominates the final part of the trajectory toward position. Here (upper right part of Fig. 4) pulse height and agonistic pulse width have been kept constant, whereas antagonistic pulse width has been changed within a range of zero to 98 ms about the nominal value of 14 ms. This display shows clearly how antagonistic pulse width is certainly less important than agonistic pulse width, but how it finely adjusts duration and final magnitude of the movement. It is most interesting to look at an antagonistic pulse width that is longer than the nominal agonistic pulse width; this movement increases to a certain amplitude and then decreases even to zero degree for a 98 ms antagonistic pulse width. Thus we see that if pulse width 2 becomes very large the movement is not only stopped, but actually reversed. However, as pulse width 2 becomes smaller and smaller an even more interesting elucidation of an important interaction between controller signals and trajectories becomes apparent. We get a prolonged positive acceleratory phase and a slow and drawn out negative acceleratory phase. This gives long slow movements such as the upper curves in the family (upper right hand part Fig. 4).

1.4. Second Agonistic Pulse width (PW3). An additional agonistic pulse (PW3) also has significant effects on the model's head position trajectory (right part Fig. 4). The nominal value of 3 ms is varied over a range from zero to 80 ms, with the other three controller signal parameters, pulse height, pulse width 1 and pulse width 2 kept constant. A second agonistic pulse (PW3) of 3 ms pulse width gives a response which is equal or close to the second order controlled response. The time optimal third order response was found to be in response to the 3 ms PW3. It appears that the time optimal second order controller response with heuristic modelling is very similar. For a larger value of PW3 the movement becomes interrupted and "double lumped". This actually occurs in experimentally recorded head rotations: e.g. in fatigued movements, and also in optionally slowed movements. The temporary braking in the middle of the triphasic position curve may be somewhat artifical here, since PW2 may have been set too high in this parametric behavioral study (see also for comparison Fig. 3). Overall one can see that the second agonistic pulse (PW3) is not so important for a time optimal movement, but plays a more significant role in slower movements. We noted that with a time constant of 50 ms for activation and deactivation the third short pulse can not generate much force; this is somewhat different from eye movements with much shorter time constants.

Generally, it is obvious how a change of pulse height should be accompanied by a change in agonistic and mostly antagonistic pulse width in order to get a realistic response. Of course the reader should note that all of these controller signal parametric adjustments are done with strict adherence to reciprocal innervation, and therefore these examples give us a realistic "practical" insight into the effect of these controller signal changes.

1.5. Controller Signal Envelopes. The range of shapes of controller signals having three orders and four amplitudes (Fig. 5) that we used in these modelling studies are most instructive in understanding the physical nature of the plant and of the strategies possible to produce the dynamic trajectories by the controller signal. Here we show the controller signals for the agonist and antagonist pulse envelopes: first order control, leftmost family of curves, second order control, middle family of curves, and third order control, rightmost family of curves. Pulse envelopes are overall functions describing the neural pulse frequency driving the head movement trajectory. The width of the pulse envelope represents the duration of the high burst of agonist or antagonistic tiring. The amplitude of the pulse envelope represents the rate of firing and also the number of neurons that are themselves firing, which of

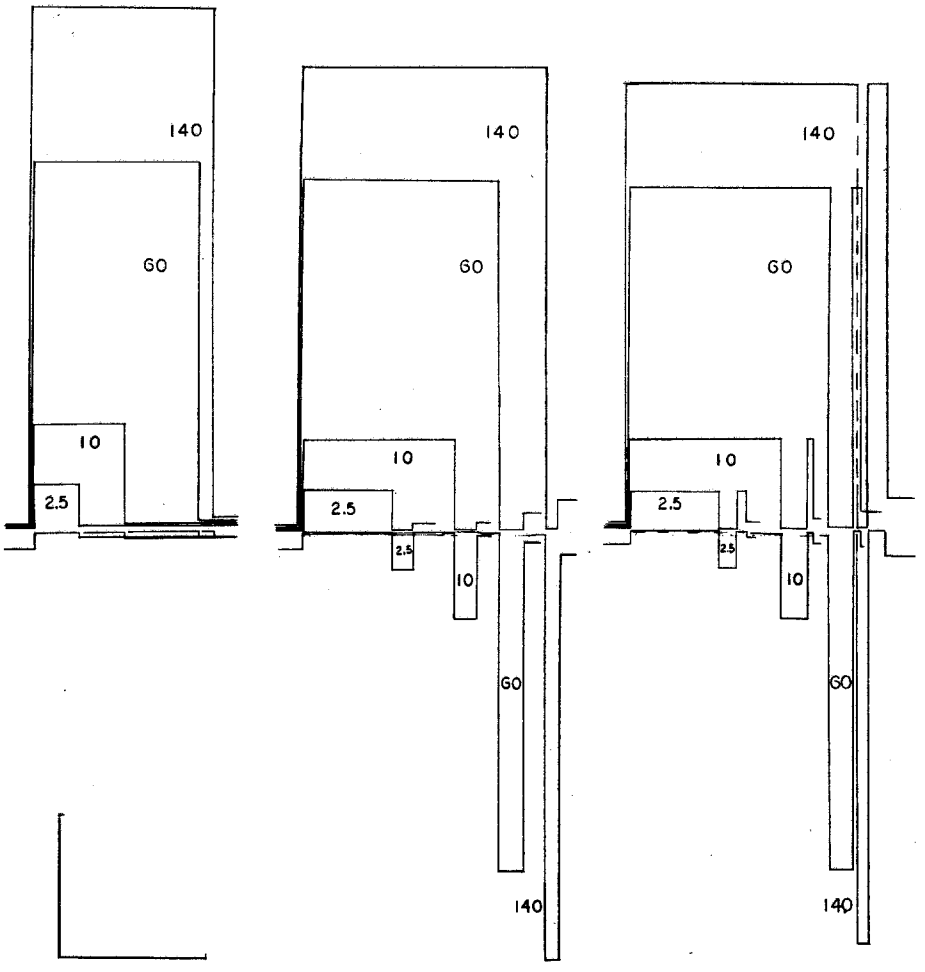


Fig. 5. Envelopes of controller signals of first, second and third order (left to right) for four amplitudes: 2.5, 10, 60, 140 degree (ordinate scale). Abscissa scale is pulse width in msec; calibration mark: 1000 gram-force, 100 ms

course goes up as more neurons are recruited (Henneman et al., 1965). The summation of individual single unit bursts, to make the pulse envelope represent a change in conceptual thinking from a single unit neuron recording of the neurophysiologist to the overall consideration of pulse firing rate and the numbers of neurons that together might be considered to be the neurological control signal. This shows the close relationship between neurological considerations and engineering control theory. If one wishes to dissect below the level of the pulse envelopes of the engineering controller signals, one then is in the realm of neurophysiology – closer to neurophysiological events, but rather away from the actual neurological control signals which are producing the behavior.

In modeling we convert pulse height in terms of numbers of pulses fired (the product of numbers of neurons times average firing rate) into hypothetical Hill tension produced in the muscle in grams of force. Thus the ordinate scale of the display is actually in terms of grams force, with, for example the 140 degree movement receiving 4200 g force of pulse height and the sixty degree movement receiving 3500 g force. The tonic force for the agonist is 16 g force, a rather low value. The antagonist curves below the agonist curves represent, for example, a change of grams force from minus 16 to minus 2. The time scale is in ms, for example the first order 140 degree trajectory is driven by a 164 ms pulse width, whereas the 60 degree amplitude head rotation trajectory is driven by a 140 ms pulse width, the difference being a 24 ms period; these numbers also give an idea of the scale of the display. Several important aspects of these controller signal shapes, their height and durations, show up at once.

When we compare the first, second and third order overall shapes, they appear somewhat similar in pulse height and pulse width. However when we look in detail we see that the pulse width is much shorter for the first order responses. This is true because there are no braking antagonistic pulses which can terminate the movement. The first order control needs a "coasting time" in order for the movement to terminate.

Within an order pulse height certainly increases in order to produce larger head movements. However, comparing between orders we see that pulse height in the lower order controller signals can be traded for

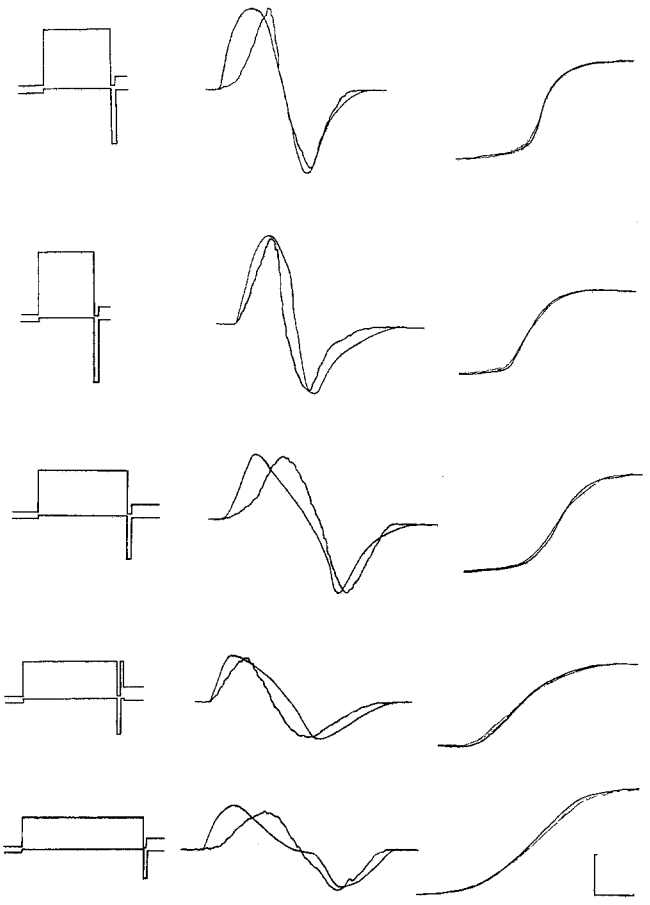


Fig. 6. Model fit for common head acceleration types. Left column: controller signal envelopes; middle column: acceleration traces (thicker continuous lines equal model trace); right column: position traces. Calibration mark: abscissa time, 100 ms. Ordinate: for left, middle, right, respectively: 1000 gram-force, 1000 degree per sec. per sec., 10 degrees.

pulse width in the higher order; thus the higher order controller signals need to employ less force, but employ this smaller force for a longer duration. These controller signals are not normalized as is usually done in such displays. Thus we can clearly see the enormous range of pulse heights of these envelopes, representing neurological control exerted by the central nervous system, to control head rotations.

The very small differences in pulse width as a function of amplitude were surprising to us in comparision to our earlier studies of eye movements; for example there is only 24 ms difference between the pulse envelopes for the 60 degree head and 140 degree trajectory generated by first order control; similarly, there is only about 50 ms difference in the pulse width controlling the 10 degree and 60 degree head rotational trajectories for second order control. This showed us why it is so difficult to use only EMG studies for estimating the controller signals. Given the inherent imprecision and sampling aspects of the EMG recordings, it is difficult, if not impossible, to observe such small differences.

The higher order controller signals are of very short duration. This has some interesting implications in the interactions of certain other parameters, such as the time constants of activation and deactivation. Since these are modelled as first order time constants it is clear that the actual activation of the muscle will be varied much less than the amplitude of the square topped pulse envelopes. Indeed, the actual range of tensions produced by these very short duration pulses is rather small and much more influenced by these time constants of activation and deactivation than by pulse height. A sensitivity analysis of the relative influence of these parameters as a function of the higher order controller signal parameters and amplitudes would demonstrate this.

2. Types of Head Model Accelerations

2.1. Common Types

Experimentally, many different types of head movement trajectories can be noted (as described above, Fig. 2, method section). In this way the head movement contrasts with the rather stereotyped saccadic eye movement. It has long been neurological wisdom that the head movement trajectory may have important influences from the stretch reflex and also have much more voluntary control than the eye movement saccade. One of the uses to which we now put the model is to study the changes in the model which simulate these different types. A sample representation of these types are arranged with: first controller signal, then model and experimental acceleration in the middle, and to the right experimental and model position change in the display (Fig. 6). The calibration mark is used for the three parts of each type with time being the abscissa and the ordinate changing from grams force (pulse frequency times numbers of neurons recruited) on the left hand side, degrees per second per second in the middle, to degrees for the right hand side. The movements were 43 degrees of horizontal head rotations. The five horizontal rows represent five different types arranged in decreasing order of maxium acceleration. As can be noted by looking at the right hand column the model fits the experimental position trajectory very well for all five types. The model also shows excellent fits to other behavioral features, such as duration, maximum velocity, maximum positive acceleration and maximum negative acceleration. The timing behavioral parameters, however, were less well fitted as can be noted in the middle column showing model and experimental acceleration trajectories. The second row

shows a second order control signal producing an acceleration maximum of 2200 degrees per second per second and an excellent position fit. The model and experimental trace represens the symmetrical acceleration type 1 and also type 2 which shows a slight asymmetry with maximum positive acceleration being larger than maximum negative acceleration. It also shows either convex or concave final return of negative acceleration to base line. This is the most common experimental type and represents a fairly consistent time optimally intended response produced by human experimental subjects intending to move their head in horizontal rotation as fast as possible.

When we look to the first row, which again shows a second order controller signal, we see a controller signal with longer duration and a somewhat decreased amplitude as compared with the one of the second row. This controller signal produces an asymmetrical type 4, in which the maximum negative acceleration is somewhat greater than the maximum positive acceleration, as can be noted in the experimental acceleration curve in the middle column. This model response shows again a very high acceleration of 2300 degrees per second per second and is still almost time optimal. It does not show any overshoot, however, inspite of the large negative acceleration. As can be clearly seen when comparing the model and experimental traces in the middle column, the model is "too hot". By this we mean, that the acceleration curve rises much sooner with the model than with the experimental acceleration trajectory. We could of course have adjusted the model to eliminate this discrepancy. However we chose not to, because our EMG evidence suggested, that the most likely change to make would have been a rather straightforward and adhoc reduction in the early onset of the first pulse.

Letting the first pulse rise slowly would simulate either a slow increase in neural discharge of the alpha motoneurons or in maximal recruitment of the motoneuronal pool, or both (Henneman et al., 1965). The square wave controller signal assumes that maximum discharge rate and maximum recruitment is reached almost instantaneously. Although this is not true for many movements, as judged by the EMG evidence, it is true for fast, close to time optimal, movements, as the example in the second row (see for comparison leftmost movement in Fig. 3). Differences of the actual experimental controller signals from those idealized in the model are not significant at the level of precision with which the model and the experiment are being compared. However, EMG evidence suggests, that there is often an "exponential rise" in EMG firing rate, mirrored by an exponential rise or a slower rise in the acceleration (and also in the derivative of acceleration trace), [Zangemeister et al. (1980)].

The third row shows a type 3 experimental response with later occuring, smaller and more rounded maximum positive and maximum negative acceleration, an experimental type seen even more commonly. This is still a second order type. It is clearly non time optimal, especially when we look at the controller signal and see that the pulse height is too low; therefore a compensating increase in pulse width has to occur. This is seen more clearly in the fourth row, which shows a slow type 3, again with the negative acceleration maximum being larger than the positive acceleration maximum; it is an even more non time optimal response, with a very low pulse height and a very long pulse width. As before, the requirement for a tradeoff between pulse height and pulse width is evident, so that the product stays roughly constant in order to achieve the 43 degree fixed magnitude of the head rotational trajectory. Notice that this is now a third order controller signal, which was found to be necessary to best model this movement. Multiple pulse control permits the total width of the controller signal to be very close to the total width of the actual trajectory.

In the bottom row, the fifth, a very slow trajectory can be seen with a long pulse width and a low pulse height together producing a slow, non time optimal type 3, and with an irregular "braking". This experimental trajectory and its model fit also illustrates a type 5 experimental response with irregularity due to "braking" pulses.

For each of the bottom three rows we see that the model decline from maximum positive acceleration to maximum negative acceleration is slower than the experimentally found decline of the acceleratory trajectories. Slow decline, however, is not seen in the simulations of the upper rows. Also, the model duration tends to be longer than the experimental duration; effects possibly related to the first order time constants of activation and deactivation that may not actually express the true dynamics involved in excitation contraction coupling or calcium activation and deactivation of muscles. Although pulse width 2 was essential in order to get time optimal fits and best fits given the constraints on our model, it did not seem to change much with modelling for different types as can be noted in the first column (Fig. 6). Similarly, the third order controller signal appears not to play a very decisive role in fitting of the different types. In only one or our model simulations, fourth row, did a third order controller signal appear to give a better fit.

2.2. Amplitude and Dynamic Overshoot

The model also was fitted to two quite different response amplitudes and to response with and without

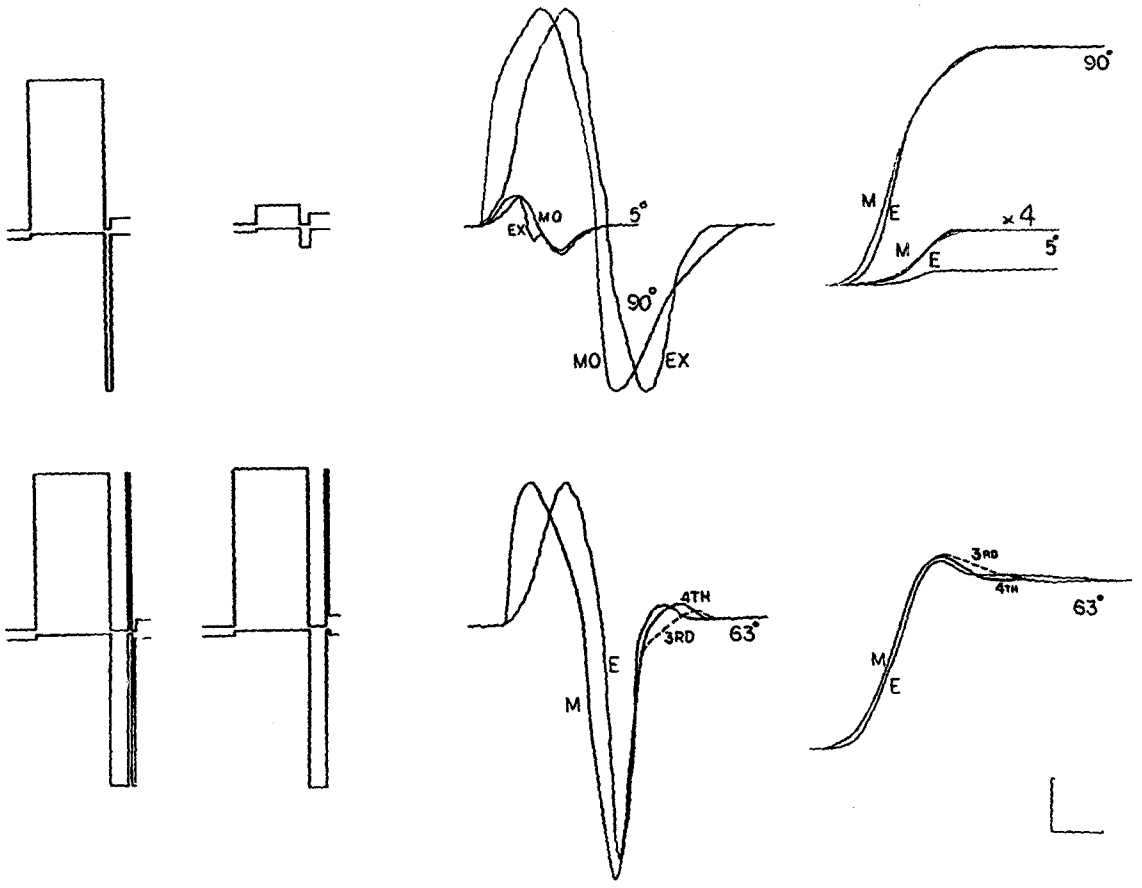


Fig. 7. Model fit for highly different amplitudes and dynamic overshoot. Left column: controller signal envelopes; middle column: acceleration traces; right column: position traces. Calibration as in Fig. 6

dynamic overshoot (Fig. 7), as an excellent test. This also enabled us to use the model to explain these interesting aspects of the control of head rotation – an important role of the model for us. The first two column of Fig. 7 represent controller signals. The rightmost column shows the position traces for model and experiment and illustrates the excellent fit of model trajectory to experimental trajectory. The middle column shows the acceleration traces where we can best demonstrate divergences of the model trajectory from experimental trajectories. Again calibration represents time in msec on the abscissa and for the ordinates grams force, acceleration, and position, respectively. The upper row shows controller signals for a 90 degree and a 5 degree head rotational trajectory. These are not the extrema of our experimental situation, but nevertheless represent an extremely large range difference. Second order controller signals are used for both fits. The right column shows excellent fit for both the 90 degree and the 5 degree movement; indeed we had to amplify the 5 degree movement in order to show even minor discrepancies between model and experimental curves. When we look at the middle column, the acceleration traces, we see again that the model is "too hot". The extent of the discrepancy between model and experiment is only apparently larger for the 90 degree movement than for the 5 degree movement – both time functions show approximately equivalent discrepancies; thus the excessive dynamical speed in the initial portions of the model trajectory is not a function of amplitude.

The bottom row (Fig. 7) shows the attempt to fit a dynamical overshoot experimental trajectory with both a fourth order (leftmost) controller signal and a third order (right) controller signal (first column). Note that the additional controller signal pulses are very narrow. Also recall that the heights of these narrow controller signal processors are not very influential parameters since the actual amplitude is more controlled by the time constants of activation and deactivation as described. It can be seen that the fourth order controller signal produces a better fit than the third order controller signals in both position and acceleration traces. The dynamic overshoot can not in fact be a second order additional response to the second order main trajectory, since the PW 3 is so much shorter than PW 1 of an equivalent size trajectory (see Fig. 4). The model maximum negative acceleration, a bit larger

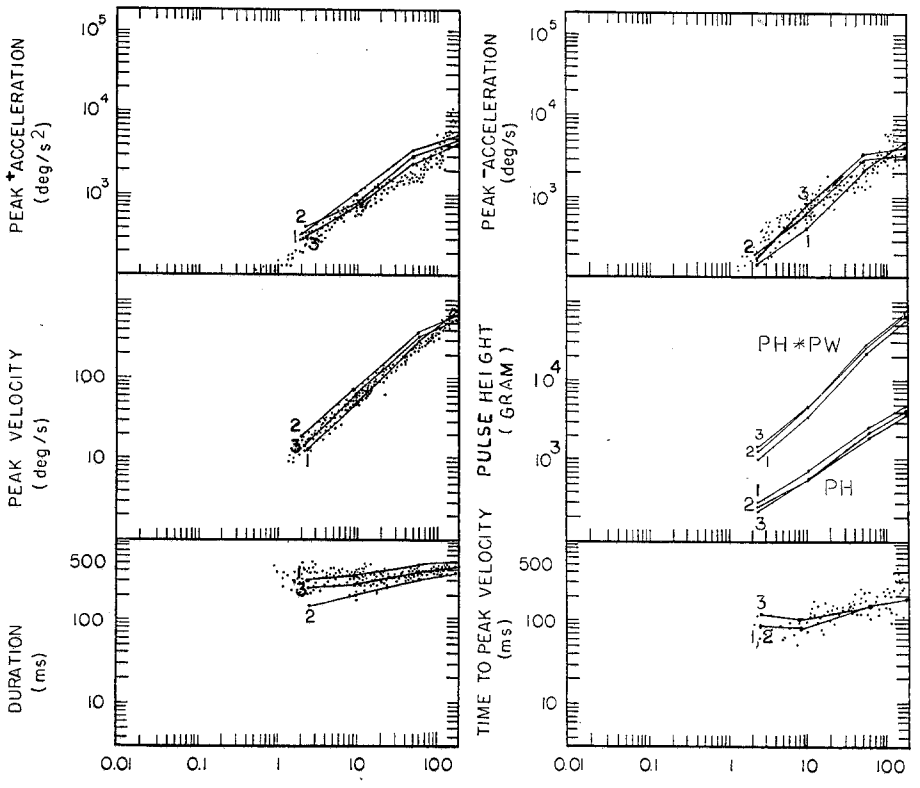


Fig. 8. Model fit for head movement main sequence. Double logarithmic scale. Abscissa magnitude in degree, ordinate as indicated. Data points are over hundred single points from one subject, that were also reconfirmed in several other subjects. 1, 2, 3, first, second, third order signals. PH, PW: pulse height, pulse width

than the experimental negative maximum acceleration, was difficult to adjust in additional adjustments.

Again the model response develops its positive acceleration too rapidly. These additional simulations also suggest that a correction using a ramp or an exponential rise of controller signal, a fairly unocuos modification of the model, would make it fit the exponential data much more closely. However, as others in the field of simulation of biological phenomena well know, the aim is not to fit model to experiment by adhod changes, but rather to expose and explicate important dynamical and neurological control phenomena underlying the experimental trajectories. These additional model simulations of the wide amplitude range and of dynamic overshoot demonstrate extremely good fit of the model to experimental data for position traces. Differences in acceleration traces provide physical insight into the dynamics of head rotational system and the neurological signals that do indeed control these head rotations.

3. Model Fit to Head Movement Main Sequence Data

The ability of the model to closely fit experimental data is also exceptionally well shown when we attempt to fit the experimental data displayed in the main sequence relationship (Fig. 8). In this extended main sequence diagram with many behavioral variables and controller signal parameters as functions of amplitude of the head horizontal rotation trajectory the points are data from some hundred experimental head rotational trajectories on one subject supported and confirmed by data from many other subjects (Zangemeister et al., 1981)]. Behavioural data from our model simulations with 1, 2, and 3 representing simulations using first, second and third order controller signals, respectively, are shown as lines, fitting the experimental data points. It is clear that all three orders of controller signals fit the experimental data quite closely.

When we consider our special interest in time optimality, that is, what controller signals will produce the fastest possible movement, we see that the shortest durations are given by the second order controller signal (lower left hand part, Fig. 8). The relative positions of the different controller signals on the diagram for peak velocities, accelerations and time to peak velocity vary in detail. For example, maximum velocity is given by second order controller signals over a fairly wide range of model trajectories, with the first order controller signal being the slowest. The third

order controller signal gives the longest time to maximum velocity for small amplitude head rotations.

When we look to pulse height as a function of trajectory magnitude we see that pulse height for the first order control is somewhat larger than for higher order controls. Another indication where first order control needs a considerably higher pulse height and then a long period to coast into stable final position, than does higher order control with its traking antagonist pulses. These questions of efficency are more complex than might seem at first glance. The product of pulse height times pulse width, which after all is the total amount of effort (actually, number of nerve impulses) expended, shows first order control to have perhaps an advantage over higher order control. Generally, the discrepancies between optimal fitting with any of the orders of controller signal are relatively slight when considering the scatter of the data; also our general notions of second order control being time optimal appear to be somewhat substantiated by these model fits of experimental data.

Acknowledgement. We are pleased to acknowledge partial support from the NCC2–86 Cooperative Agreement, NASA-Ames Research Center, and from the National Eye Institute Training Grant EYO7043.

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Received: January 30, 1981

Dr. W. H. Zangemeister
Neurologische Universitätsklinik
Martinistrasse 52
D-2000 Hamburg 20
Federal Republic of Germany